CS4045 — Deep Learning for Perception
Project Guidelines

Fall 2026 | 10 grade points | Proposal Submission Week 9 | Final presentation: Week 16

1. Project overview

Develop a small, working deep-learning application that addresses a clearly defined computer-vision problem. The project should demonstrate that you can apply the CNN, object-detection (YOLO) and other concepts covered in class, use a deep-learning framework, evaluate model outputs, and explain limitations. Each group must preferably select ONE of the two project routes below.

The two routes differ in the prediction task and data preparation. Image classification predicts a label for an entire image. Object detection predicts object classes and their locations using bounding boxes. Choose the route based on the problem, available data, and time; neither route is automatically worth more marks.

[We may decide to add more options – These options are based on the course covered so far]

2. Choose one option

Option A — Image classification

Use a pretrained classifier (e.g., MobileNetV2) to assign a category to an entire image. Example: classify campus images as “litter present” or “no visible litter.” Label each image; bounding boxes are not needed. Evaluate with suitable metrics such as accuracy, precision, recall, F1-score, and a confusion matrix.

Option B — Object detection

Use pretrained YOLO to identify objects and their locations. Example: detect litter in campus images. Annotate objects with bounding boxes, run a pretrained baseline, then fine-tune YOLO on the labelled dataset. Evaluate with precision, recall, and mAP, and show examples of correct and incorrect detections.

3. Proposal — due by the end of Week 9

Submit a concise proposal containing:

- Project title and selected option.
- All group members’ names, student IDs, university emails, and group coordinator.
- Problem statement, intended use, and proposed classes.
- Dataset source/collection plan, estimated size, and permission/privacy considerations.
- Pretrained model, baseline, evaluation metrics, and planned timeline.
- Task allocation and expected contribution of each member.

Topic and scope may require instructor/TA approval.

4. Dataset and ethical requirements

- Collect your own images where practical and permitted. If not feasible, use an instructor-approved public dataset. In the case of the public dataset, you will still be required to test on hand collected and engineered data. You can compare the performance on both publicly available data and your own collected data.
- Record the data source, licence/permission, and who collected or annotated the data. Public availability does not automatically permit unrestricted reuse or redistribution.

- Obtain campus/site permission where required. Avoid identifiable faces, number plates, ID cards, screens, private documents, and sensitive locations.
- Do not upload sensitive images to public annotation platforms or AI tools. Check privacy settings and service terms first.
- Store data securely and share it only where permitted. Follow dataset licences and cite external data, models, code, and libraries.

5. AI-assisted coding

- AI tools may help with coding and debugging subject to university policy.
- Verify generated code if any and understand the main data-processing, model, training, and evaluation steps.
- Briefly disclose which AI tools were used and for what purpose.
- Each student must be able to explain their contribution and answer technical questions during an individual oral defense. You should be able to explain every bit of your code.

6. Workflow and timeline

When	Milestone
By end of Week 9	Submit group proposal for approval.
Week 16	Submit project and deliver presentation.

7. Deliverables

- Runnable notebook/script and README with run instructions.
- Dataset description and source.
- Baseline and final results, appropriate metrics, and examples of errors.
- Report of approximately 8-10 pages covering method, results, limitations, ethics, and AI-use disclosure.
- 10-minute presentation/demo and individual oral defense.

8. Assessment rubric (10 grade points)

Criterion	Points (50), weightage = 10 marks
Problem definition and proposal	5
Novelty, Real life application of the project	5
Dataset quality, provenance, and labels/annotations	5
Implementation and pretrained model use	10
Performance Evaluation and error Analysis	10
Technical understanding and individual defense (Presentation)	10
Report, reproducibility, ethics, and AI disclosure	5

9. Suggested resources

- **Keras MobileNetV2:** https://keras.io/api/applications/mobilenet/
- **Ultralytics YOLO prediction:** https://docs.ultralytics.com/modes/predict
- **Ultralytics YOLO training:** https://docs.ultralytics.com/modes/train
- **Roboflow annotation:** https://roboflow.com/
- **Google Colab:** https://colab.research.google.com/

More instruction may be furnished later on.